



A distance-based statistical analysis of fuzzy number-valued data



A. Blanco-Fernández, M.R. Casals, A. Colubi, N. Corral, M. García-Bárcana,
M.A. Gil, G. González-Rodríguez, M.T. López, M.A. Lubiano, M. Montenegro,
A.B. Ramos-Guajardo, S. de la Rosa de Saa, B. Sinova

SMIRE Research Group at the University of Oviedo ^{*,1}

Dpto. de Estadística e I.O. y D.M., Universidad de Oviedo, 33007 Oviedo, Spain

ARTICLE INFO

Article history:
Available online 16 October 2013

Keywords:
Fuzzy data
Random experiments
Distance between fuzzy data
Statistics

ABSTRACT

Real-life data associated with experimental outcomes are not always real-valued. In particular, opinions, perceptions, ratings, etc., are often assumed to be vague in nature, especially when they come from human valuations. Fuzzy numbers have extensively been considered to provide us with a convenient tool to express these vague data. In analyzing fuzzy data from a statistical perspective one finds two key obstacles, namely, the nonlinearity associated with the usual arithmetic with fuzzy data and the lack of suitable models and limit results for the distribution of fuzzy-valued statistics. These obstacles can be frequently bypassed by using an appropriate metric between fuzzy data, the notion of random fuzzy set and a bootstrapped central limit theorem for general space-valued random elements. This paper aims to review these ideas and a methodology for the statistical analysis of fuzzy number data which has been developed along the last years.

© 2013 Published by Elsevier Inc.

1. Introduction

Traditionally, most of the statistical developments refer to the analysis of real-valued data. However, in the real world other types of data often arise, so that Statistics is being enlarged by considering new ideas, concepts and results. In this respect, one can mention vector-valued, censored, set-valued, spatial, and functional data, among others, as well as their corresponding analyses, to cover a wide diversity of data that one can often find in real-life.

On the other hand, in the last decades fuzzy sets have been shown to be a powerful tool to represent quantitatively (more precisely, functionally) many imprecise real-life data. More concretely, we often find situations in which the values of some involved variables or attributes can only be observed imprecisely. For instance, when opinions, quality ratings, satisfaction valuations and many other surveys are conducted, responses that usually vary from respondent to respondent and from question to question (that is, they are subject to randomness) cannot be expected to be expressible in terms of values for a precise scale.

The approach followed in this paper is the so-called ‘ontic’ one (see, for instance, Dubois and Prade [13]), where ontic means to pertain to the essence of the object under study. Three main situations support this approach:

* Corresponding author. Tel.: +34 985103356; fax: +34 985103354.

¹ E-mail address: SMIRE@uniovi.es.

- When the observer imprecisely measures a precisely defined attribute but we study the imprecise data he/she provides (fuzzy numbers are ontic w.r.t. the observer and they describe how much the observer is imprecise);
- When perceptions are collected from a human but there is no objective value behind (e.g., the taste of a dessert does not directly describe the objective ingredients of the dish); here the ontic view refers to the human tasting;
- When an intrinsically gradual representation subject to variability in the nature is described by means of fuzzy sets (e.g., regions with gradual boundaries in an image, flexible time intervals, a vector of ratings, etc.).

As we will later emphasize, the ontic approach is to be treated by considering Féron/Puri and Ralescu's fuzzy-valued random elements (see [14,46]). These random elements associate a fuzzy value with each experimental outcome and related statistical methods attempt to draw conclusions about them. In contrast, a different view is offered by Kruse and Meyer's fuzzy random variables [37] that formalize the fuzzy perception of an underlying (original) real-valued random variable (i.e., the 'epistemic' approach is followed, where epistemic means to pertain to the quality of the knowledge that can be obtained on the object under study). A fuzzy random variable in Kruse and Meyer's sense made reference to the set of possible originals and statistical methods attempt to draw conclusions about the unknown real-valued original.

To treat fuzzy number data with statistical purposes some peculiarities should be pointed out:

- If one tries to perform a descriptive study summarizing the information contained in a sample of fuzzy data, then one should basically focus on the arithmetic, the ranking and the distances between these data. At this point, two main drawbacks can be stressed in developing descriptive statistics with fuzzy number data, namely,
 - The usual and most natural arithmetic between fuzzy numbers extends the one between real values and the usual one between intervals. It involves special features in contrast to the real-valued case, due to the lack of linearity associated with it. As a consequence, there is no definition for the difference of fuzzy numbers which is simultaneously well-defined and preserves some essential properties of the difference in the real-valued case, whence the use of the difference should be mostly avoided;
 - A universally acceptable complete order between fuzzy numbers cannot be established, and therefore the notion of empirical cumulative distribution function and related summary measures cannot be formalized in the usual way.
- To overcome these drawbacks, one possible/common solution is to define a proper distance between fuzzy numbers. A suitable metric between fuzzy numbers can be employed to quantify the average variability of data, the error associated with the approximation of some measures or relationships, etc. In this way, one can cope with the formalization and computation of some summary measures of the fuzzy datasets, and with the development of the descriptive fitting of some functional relationships (i.e., regression analyses) when two- or multi-dimensional fuzzy data are analyzed.
- If one tries to perform an inferential study about a random mechanism generating fuzzy number-valued data (in which fuzziness affects data values, whereas randomness affects their generation) on the basis of a sample from it, then one should mainly focus on defining suitable statistics and obtaining their sampling distributions. At this point, in addition to the drawbacks already mentioned for the descriptive analysis, two new concerns can be noticed in developing inferential statistics with fuzzy number data, namely,
 - There are not yet probability distribution models which have been shown and proved to be convenient and realistic enough to deal with the random mechanisms producing fuzzy number data;
 - There are not Central Limit Theorems for random mechanisms producing fuzzy data which could be directly applied in inferential developments because it ensures that the limit element takes values on the space of fuzzy numbers.

To overcome these drawbacks, three tools will be crucial: the notion of random fuzzy number and its induced distribution; the real-valued distance-based measures or statistics (quantifying either the error in the fuzzy number-valued estimation or the deviation between the reality of the sample fuzzy information and a hypothetical model in testing hypothesis); the existence of a bootstrapped Central Limit Theorem for generalized space-valued random elements allowing us to apply re-sampling techniques which guarantee the well-definition of the involved elements as fuzzy numbers. In this way, one can cope with establishing procedures either to estimate some 'parameters' of the distribution of random mechanisms or to test some statistical hypotheses about them, by preserving the essential ideas, concepts and approaches used to solve these problems in the real-valued case.

This paper aims to describe the main ideas and results on a distance-based methodology for the statistical analysis of fuzzy number data which has been developed along the last years. In Section 2, the considered metric and some metric properties will be recalled; some equivalent expressions will also be given, due to the practical and theoretical implications derived from such equivalences. Section 3 will be devoted to the formalization of the random mechanisms producing fuzzy number data (random fuzzy numbers), their connection with the distance and some relevant distance-based summary measures. Section 4 will examine the distance-based methodology to analyze fuzzy number data. The paper will conclude with some comments related to an application of the preceding methodology to develop inferential analysis on the distributions of real-valued random variables.

The concepts and methods that appear in this review, along with other ones which are not based on the same metric, are described in detail in the original manuscripts that introduce them (references will be given in due course) as well as, to some extent, in some recent reviews prepared by invitation (see Blanco-Fernández et al. [2], Colubi et al. [3] and Gil et al. [19]). These reviews refer to the more general case of fuzzy-valued data and random fuzzy sets indeed. The review in

this paper aims at covering the most frequent, applicable and ease-to-interpret situations dealing with fuzzy number data formalized in terms of random fuzzy numbers. The paper adds a touch of history on the key fundamentals and tools.

2. Bertoluzza et al.'s metric: definition and equivalent expressions

Although several metrics between fuzzy sets have been introduced in the literature (see, for instance, Klement et al. [33], Diamond and Kloeden [10]) the one to be considered in this paper is a more recent and versatile one which is quite well-adapted for many developments with fuzzy number data.

Let $\mathcal{F}_c(\mathbb{R})$ denote the class of fuzzy numbers $\tilde{U} : \mathbb{R} \rightarrow [0, 1]$ for which the α -levels $\tilde{U}_\alpha$ are nonempty compact intervals for all $\alpha \in (0, 1]$, where $\tilde{U}_\alpha = \{x \in \mathbb{R} : \tilde{U}(x) \geq \alpha\}$. Some authors refer to $\mathcal{F}_c(\mathbb{R})$ as the class of the fuzzy intervals.

It is quite rational for a distance between fuzzy numbers to extend, on one hand, the Euclidean metric between real values and, on the other hand and level-wise, a suitable distance between intervals.

In this way, the (generalized) metric introduced by Bertoluzza et al. [1] is defined as it follows. If

- W is a normalized weighting measure on the measurable space $([0, 1], \mathcal{B}_{[0,1]})$ which is formalized as a probability measure associated with a non-degenerate distribution,
- and φ is a normalized weighting measure on the measurable space $([0, 1], \mathcal{B}_{[0,1]})$ which is formalized as a probability measure associated with an absolutely continuous distribution function being strictly increasing on $(0, 1]$,

then

Definition 2.1. The (W, φ) -distance between two fuzzy numbers $\tilde{U}, \tilde{V} \in \mathcal{F}_c(\mathbb{R})$ such that $l_{\tilde{U}}, r_{\tilde{U}}, l_{\tilde{V}}, r_{\tilde{V}} \in L^2((0, 1], \mathcal{B}_{(0,1]}, \varphi)$ (with $L^2((0, 1], \mathcal{B}_{(0,1]}, \varphi) = \{\text{squared integrable functions w.r.t. the probability space } ((0, 1], \mathcal{B}_{(0,1]}, \varphi)\}$, and $l_{\tilde{U}}(\alpha) = \inf \tilde{U}_\alpha$, $r_{\tilde{U}}(\alpha) = \sup \tilde{U}_\alpha$), is given by the value

$$D_W^\varphi(\tilde{U}, \tilde{V}) = \sqrt{\int_{(0,1]} [d_W(\tilde{U}_\alpha, \tilde{V}_\alpha)]^2 d\varphi(\alpha)} \quad (1)$$

where

$$d_W(\tilde{U}_\alpha, \tilde{V}_\alpha) = \sqrt{\int_{[0,1]} [\tilde{U}_\alpha^{[t]} - \tilde{V}_\alpha^{[t]}]^2 dW(t)}$$

and $\tilde{U}_\alpha^{[t]} = t \cdot \sup \tilde{U}_\alpha + (1 - t) \cdot \inf \tilde{U}_\alpha$ (i.e., $(\alpha, \tilde{U}_\alpha^{[t]})$ is a so-called gradual number, see Dubois and Prade [12]).

The (W, φ) -distance in Eq. (1) is an L^2 -type metric and if the space of fuzzy numbers for which the distance is well-defined, $\mathcal{F}_c^\varphi(\mathbb{R}) = \{\tilde{U} \in \mathcal{F}_c(\mathbb{R}) : l_{\tilde{U}}, r_{\tilde{U}} \in L^2((0, 1], \mathcal{B}_{(0,1]}, \varphi)\}$, is endowed with it, a separable metric space is obtained (see, for instance, [1,26,50]). Several comments should be emphasized in connection with the metric.

Remark 2.1. To interpret the metric D_W^φ intuitively, one can notice that to quantify the distance between $\tilde{U}$ and $\tilde{V}$:

- a double bijection between $\tilde{U}$ and $\tilde{V}$ is first established by associating
 - for any arbitrary $\alpha \in [0, 1]$: $\tilde{U}_\alpha \leftrightarrow \tilde{V}_\alpha$,
 - and for any arbitrary $t \in [0, 1]$: $\tilde{U}_\alpha^{[t]} \leftrightarrow \tilde{V}_\alpha^{[t]}$;
- the root mean square Euclidean distance between the points associated through the double bijection (see Fig. 1), that is, $[\tilde{U}_\alpha^{[t]} - \tilde{V}_\alpha^{[t]}]^2$ is later computed.

It should be pointed out that in the definition of the (W, φ) -distance W and φ have not stochastic but weighting meaning.

The choice of φ allows us to weight the influence of each level (i.e., degree of ‘vagueness’, ‘fuzziness’, ...). In this way, the choice of $\varphi = \text{Lebesgue measure on } (0, 1]$ determines a metric that is mainly sensitive to ‘location’ changes. In many respects, this selection seems to be more objective than that of φ 's for which either the lower the degree of vagueness, the higher the weight (and hence they are very sensitive to changes at the lowest levels of compatibility) or the higher the degree of vagueness, the lower the weight (and hence being very sensitive to changes at the highest levels of compatibility).

The metric D_W^φ can alternatively be expressed in some different ways. Certainly, the one in Definition 2.1 allows us to make the intuition motivating the distance in terms of the choice of the weighting measure W easier to grasp. However, some equivalences have been shown to be valuable for empirical developments as well as for some theoretical studies. These equivalences are the following:

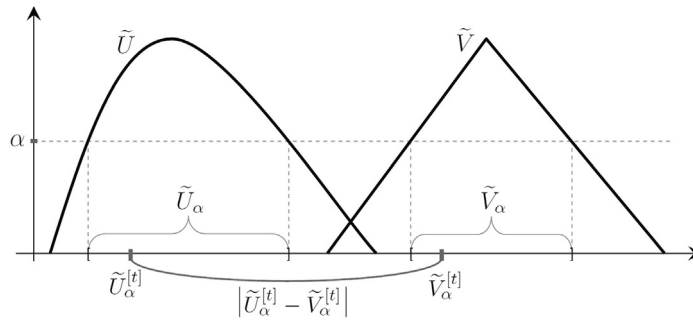


Fig. 1. The (W, φ) -distance is a root mean square distance, the distance being the one which is graphically displayed, $|\tilde{U}_\alpha^{[t]} - \tilde{V}_\alpha^{[t]}|$.

I. D_W^φ can be fully characterized (see Bertoluzza et al. [1], Lubiano et al. [40]), by particularizing the general weighting measure W to a discrete one weighting (for each level) three points: the two extremes and an intermediate one (usually the mid-point). Thus,

$$D_W^\varphi(\tilde{U}, \tilde{V}) = D_\lambda^\varphi(\tilde{U}, \tilde{V}) = D_{(\lambda_1, \lambda_2, \lambda_3)}^\varphi(\tilde{U}, \tilde{V}) = \sqrt{\int_{(0,1)} (\lambda_1 [\tilde{U}_\alpha^{[1]} - \tilde{V}_\alpha^{[1]}]^2 + \lambda_2 [\tilde{U}_\alpha^{[t_W]} - \tilde{V}_\alpha^{[t_W]}]^2 + \lambda_3 [\tilde{U}_\alpha^{[0]} - \tilde{V}_\alpha^{[0]}]^2) d\varphi(\alpha)}, \quad (2)$$

with

$$\lambda_1 = \frac{\int_{[0,1]} (t - t_W)^2 dW(t)}{1 - t_W}, \quad \lambda_2 = \frac{\int_{[0,1]} t(1 - t) dW(t)}{t_W(1 - t_W)}, \quad \lambda_3 = \frac{\int_{[0,1]} (t - t_W)^2 dW(t)}{t_W},$$

whence $\lambda_1 > 0$, $\lambda_2 \geq 0$, $\lambda_3 > 0$, $\lambda_1 + \lambda_2 + \lambda_3 = 1$, and $t_W = \int_{[0,1]} t dW(t)$. If $t_W = .5$, then $\tilde{U}_\alpha^{[t_W]} = \text{mid } \tilde{U}_\alpha = \text{centre of } \tilde{U}_\alpha = (\inf \tilde{U}_\alpha + \sup \tilde{U}_\alpha)/2$.

In summary, although Eq. (2) can be viewed as a special case of Eq. (1), both represent the same family of distances. As we have mentioned, it is often more intuitive choosing W than choosing the weighting vector $\lambda = (\lambda_1, \lambda_2, \lambda_3)$, but the last one is easier to handle in simulations.

II. D_W^φ can also be characterized (see Gil et al. [20], Trutschnig et al. [50]) by replacing the general weighting measure W by a non-stochastic weighting $(1, \theta_W)$ of the squared distances between some intermediate points and the squared distances between the spreads as follows:

$$D_W^\varphi(\tilde{U}, \tilde{V}) = \sqrt{\int_{[0,1]} \left([\tilde{U}_\alpha^{[t_W]} - \tilde{V}_\alpha^{[t_W]}]^2 + \theta_W \int_{[0,1]} [\text{spr } \tilde{U}_\alpha - \text{spr } \tilde{V}_\alpha]^2 \right) d\varphi(\alpha)}$$

where $\text{spr } \tilde{U}_\alpha = \text{radius or semi-amplitude of } \tilde{U}_\alpha = (\sup \tilde{U}_\alpha - \inf \tilde{U}_\alpha)/2$ and $\theta_W = 4 \int_{[0,1]} (t - t_W)^2 dW(t) \in (0, 1]$. Conversely (see Gil et al. [20], Trutschnig et al. [50]), for any $\theta \in (0, 1]$, there exists a weighting measure W associated with a non-degenerate probability measure on $([0, 1], \mathcal{B}_{[0,1]})$ (actually, $t_W = .5$ and $\theta = \int_{[0,1]} (2t - 1)^2 dW(t)$), such that

$$D_W^\varphi(\tilde{U}, \tilde{V}) = D_\theta^\varphi(\tilde{U}, \tilde{V}) = \sqrt{\int_{[0,1]} \left([\text{mid } \tilde{U}_\alpha - \text{mid } \tilde{V}_\alpha]^2 + \theta \int_{[0,1]} [\text{spr } \tilde{U}_\alpha - \text{spr } \tilde{V}_\alpha]^2 \right) d\varphi(\alpha)}, \quad (3)$$

which can be extended for $\theta \in (0, \infty)$, and being referred to as the (θ, φ) -distance. The last equivalence entails that, for each α , the choice of W allows us to weight the effect of the distance between the spreads or radii of the α -levels (i.e., the effect of the deviation in 'shape/vagueness') in contrast to the effect of the distance between their mid-points (i.e., the effect of the deviation in 'location').

For some comparative studies, and specially to reduce the possible effect of the range of the involved fuzzy numbers, a 'standardized' metric $D_\tau^\varphi = D_\theta^\varphi/(\theta + 1)$, with $\tau = \theta/(\theta + 1) \in (0, 1)$ has also been considered. With the notation in Eq. (3), choosing $\theta = 1$ (i.e. $\tau = .5$) is equivalent to weighting only and uniformly the two squared Euclidean distances between the extreme points of the level sets (i.e., $W(\{0\}) = W(\{1\}) = .5$), so that

$$D_1^\varphi(\tilde{U}, \tilde{V}) = \sqrt{\int_{(0,1)} \left(\frac{1}{2} [\inf \tilde{U}_\alpha - \inf \tilde{V}_\alpha]^2 + \frac{1}{2} [\sup \tilde{U}_\alpha - \sup \tilde{V}_\alpha]^2 \right) d\varphi(\alpha)},$$

whereas choosing $\theta = 1/3$ (i.e., $\tau = .25$) is equivalent to weighting uniformly all the squared Euclidean distances between the convex linear extreme points of the level sets, so that

$$D_{1/3}^\varphi(\tilde{U}, \tilde{V}) = \sqrt{\int_{(0,1]} \left(\int_{[0,1]} [\tilde{U}_\alpha^{[t]} - \tilde{V}_\alpha^{[t]}]^2 dt \right) d\varphi(\alpha)}.$$

In summary, the family of distances in Eq. (1) with $t_W = \text{mid-point}$ and the family of those in Eq. (3) represent the same family. Despite of the fact that choosing W could be more intuitive than choosing θ , the choice of a single weight is easy to handle in many empirical developments and many theoretical results can be better reached. Furthermore, the characterization of the metric by means of the mid/spread representation allows a well-formalized extension to the case of fuzzy values of higher dimension (see Trutschnig et al. [50]).

III. D_W^φ can be alternatively characterized (see Näther [43] and Körner and Näther [36]) by the inner product associated with an L^2 -distance on the space of the Lebesgue integrable functions on $\{-1, 1\} \times (0, 1]$ w.r.t. the definite positive and symmetric kernel K_W such that

$$\begin{aligned} D_W^\varphi(\tilde{U}, \tilde{V}) &= D_{K_W}^\varphi(\tilde{U}, \tilde{V}) = \sqrt{\langle s_{\tilde{U}} - s_{\tilde{V}}, s_{\tilde{U}} - s_{\tilde{V}} \rangle_{K_W}} \\ &= \sqrt{\int_{((-1,1) \times (0,1])^2} (s_{\tilde{U}}(u, \alpha) - s_{\tilde{V}}(u, \alpha))(s_{\tilde{U}}(v, \beta) - s_{\tilde{V}}(v, \beta)) dK_W(u, v, \alpha, \beta)}, \end{aligned}$$

where the support function of $\tilde{U} \in \mathcal{F}_c(\mathbb{R})$ is given by $s_{\tilde{U}} : \{-1, 1\} \times (0, 1] \rightarrow \mathbb{R}$ with

$$\begin{aligned} s_{\tilde{U}}(1, \alpha) &= \sup \tilde{U}_\alpha, & s_{\tilde{U}}(-1, \alpha) &= -\inf \tilde{U}_\alpha, \\ dK_W(u, v, \alpha, \beta) &= \begin{cases} K_W^0(u, v) d\varphi(\alpha) & \text{if } \beta = \alpha, \\ 0 & \text{otherwise,} \end{cases} \\ K_W^0(1, 1) &= \int_{[0,1]} t^2 dW(t), & K_W^0(-1, -1) &= \int_{[0,1]} (1-t)^2 dW(t), \\ K_W^0(1, -1) &= K_W^0(-1, 1) = \int_{[0,1]} t(1-t) dW(t). \end{aligned}$$

From a historical perspective, D_W^φ and D_λ^φ have been the first considered versions of the metric we have just recalled, and they have been applied in the first statistical developments of the methodology in Section 4.

For the most recent developments D_K^φ , and especially D_θ^φ , have been preferred in analyzing fuzzy number data. Some of the arguments supporting such a preference are the following:

- The mid/spread representation of fuzzy numbers yields valuable results, especially in connection with regression studies (see, for instance, González-Rodríguez et al. [23]);
- D_K^φ and D_θ^φ can be extended to fuzzy sets of Euclidean spaces of higher dimension (see Körner [34], Näther [43], Körner and Näther [36] for the extension of D_K^φ , and Trutschnig et al. [50] for the extension of D_θ^φ) on the basis of the support function of fuzzy sets (Puri and Ralescu [45]), which is an alternative characterization of fuzzy sets with compact levels through their boundaries;
- The covariance of two random mechanisms producing fuzzy data can be formalized prompted by ideas for generalized space-valued random elements.

Hereinafter, metric D_θ^φ will be considered when presenting most of the other main concepts and methods. One should notice that most of the relevant positive features of D_θ^φ are still present when it is combined with the usual arithmetic between fuzzy numbers. In accordance with this arithmetic,

Definition 2.2. Given two fuzzy numbers $\tilde{U}, \tilde{V} \in \mathcal{F}_c(\mathbb{R})$, the sum of $\tilde{U}$ and $\tilde{V}$ is defined (Zadeh [51]) as the fuzzy number $\tilde{U} + \tilde{V} \in \mathcal{F}_c(\mathbb{R})$ such that

$$(\tilde{U} + \tilde{V})(t) = \sup_{(y,z) \in \mathbb{R} \times \mathbb{R}: y+z=t} \min\{\tilde{U}(y), \tilde{V}(z)\},$$

and the product of $\tilde{U}$ by the scalar $\gamma \in \mathbb{R}$ is defined (Zadeh [51]) as the fuzzy value $\gamma \cdot \tilde{U} \in \mathcal{F}_c(\mathbb{R})$ such that

$$(\gamma \cdot \tilde{U})(t) = \sup_{y \in \mathbb{R}: y=\gamma t} \tilde{U}(y) = \begin{cases} \tilde{U}(\frac{t}{\gamma}) & \text{if } \gamma \neq 0, \\ \mathbf{1}_{\{0\}}(t) & \text{if } \gamma = 0 \end{cases}$$

or, equivalently (see Nguyen [44]), for each $\alpha \in (0, 1]$:

$$\begin{aligned} (\tilde{U} + \tilde{V})_\alpha &= \text{Minkowski sum of } \tilde{U}_\alpha \text{ and } \tilde{V}_\alpha \\ &= [\inf \tilde{U}_\alpha + \inf \tilde{V}_\alpha, \sup \tilde{U}_\alpha + \sup \tilde{V}_\alpha], \end{aligned} \quad (4)$$

$$(\gamma \cdot \tilde{U})_\alpha = \gamma \cdot \tilde{U}_\alpha = \begin{cases} [\gamma \cdot \inf \tilde{U}_\alpha, \gamma \cdot \sup \tilde{U}_\alpha] & \text{if } \gamma \geq 0, \\ [\gamma \cdot \sup \tilde{U}_\alpha, \gamma \cdot \inf \tilde{U}_\alpha] & \text{otherwise,} \end{cases} \quad (5)$$

which corresponds to consider level-wise the natural interval arithmetic.

It should be pointed out that when we consider the arithmetic above, the metric D_θ^φ is translational and rotational invariant, that is, $D_\theta^\varphi(\tilde{U} + \tilde{W}, \tilde{V} + \tilde{W}) = D_\theta^\varphi(\tilde{U}, \tilde{V})$, $D_\theta^\varphi((-1) \cdot \tilde{U}, (-1) \cdot \tilde{V}) = D_\theta^\varphi(\tilde{U}, \tilde{V})$. Actually, D_W^φ satisfies the rotational invariance for all $\tilde{U}, \tilde{V} \in \mathcal{F}_c^\circ(\mathbb{R})$ if and only if $t_W = .5$.

In most of the statistical problems to be dealt with, it seems quite plausible to assume that W is symmetric in $(0, 1]$ so that the same weights are assigned to points in each level which are equidistant to the mid one, .5. Should this be the case, $t_W = .5$.

Remark 2.2. The support function $s : \mathcal{F}_c(\mathbb{R}) \rightarrow \mathcal{L}(\{-1, 1\} \times (0, 1])$, with $\mathcal{L}(\{-1, 1\} \times (0, 1]) =$ space of the real-valued Lebesgue integrable functions defined on $\{-1, 1\} \times (0, 1]$, such that $\tilde{U} \mapsto s_{\tilde{U}}$ determines an isometrical embedding of $\mathcal{F}_c^\circ(\mathbb{R})$ with the usual arithmetic and the metric D_θ^φ into a closed convex cone of the Hilbert space $\mathcal{L}(\{-1, 1\} \times (0, 1])$ with the functional arithmetic and the distance based on $\|\cdot\|_\theta^\varphi$ where

$$\|f\|_\theta^\varphi = \sqrt{\int_{(0,1]} \left(\left[\frac{f(1, \alpha) - f(-1, \alpha)}{2} \right]^2 + \theta \left[\frac{f(1, \alpha) + f(-1, \alpha)}{2} \right]^2 \right) d\varphi(\alpha)}. \quad (6)$$

The preceding embedding allows us to unequivocally identify any $\tilde{U} \in \mathcal{F}_c^\circ(\mathbb{R})$ with the function $s_{\tilde{U}} \in \mathcal{L}(\{-1, 1\} \times (0, 1])$. Consequently, fuzzy number data with the fuzzy arithmetic and the metric D_θ^φ can be translated into functional data with the functional arithmetic and the metric based on the norm $\|\cdot\|_\theta^\varphi$, and several developments and results for functional data can be applied to fuzzy data by using the appropriate identifications and correspondences. However, special care should be paid to guarantee that the formal developments are all carried out within the closed convex cone $s(\mathcal{F}_c^\circ(\mathbb{R}))$.

3. Random fuzzy numbers: definition, distribution and relevant parameters

When random experiments lead to imprecise data which can be properly described by using fuzzy values, the mechanisms generating such elements can be treated as fuzzy-valued random elements on metric spaces in accordance with Fréchet's approach. A random element in Fréchet's sense [18] is a Borel-measurable function w.r.t. the measurable space associated with the random experiment we are considering and a measurable space associated with the metric space the random element takes values on. Random elements were thought not only to deal with the best known metric spaces, but also with spaces of abstract elements associated with attributes with a non-necessarily measurable or observable nature. As instances of these attributes Fréchet [18] foresaw some intrinsically imprecise ones like morality, the state of the political spirit, the beauty impression, and so on.

With the purpose of formalizing random elements taking on fuzzy values, and by specifying the ideas by Féron [14], Puri and Ralescu [46] introduced the concept of *random fuzzy set* that they coined as fuzzy random variables. In this way, Féron [14–16] introduced the notion of random fuzzy set to model a random mechanism generating fuzzy set values. Two different definitions were proposed, namely,

- The definition formalizing random fuzzy sets as random elements taking on values on spaces of fuzzy sets endowed with certain Borel σ -fields (i.e., by following Fréchet's theory);
- The definition formalizing random fuzzy sets as extending level-wise the notion of random sets.

Consequently, in modeling random elements taking on fuzzy values a crucial role will be played by the choice of the metric to be considered either directly, or level-wise, or through the corresponding isometrical embedding.

Puri and Ralescu [45,46], by specifying the metrics and the associated Borel σ -fields in Féron's approach, formalized the notion of random fuzzy sets in a sound way within a probabilistic setting. In case of general fuzzy values one can find a recent review in Gil et al. [19]. When fuzzy values correspond to fuzzy numbers we can refer to random fuzzy numbers, and they can be established equivalently by considering any of the two definitions by Féron as follows:

Definition 3.1. Let $(\Omega, \mathcal{A}, P)$ be a probability space. A random fuzzy number associated with $(\Omega, \mathcal{A}, P)$ is a mapping $\mathcal{X} : \Omega \rightarrow \mathcal{F}_c^\circ(\mathbb{R})$ which is a measurable function with respect to the Borel σ -field generated by the topology induced by the

metric D_θ^ϕ . Equivalently, a random fuzzy number associated with $(\Omega, \mathcal{A}, P)$ is a mapping $\mathcal{X} : \Omega \rightarrow \mathcal{F}_c^\circ(\mathbb{R})$ such that for each $\alpha \in (0, 1]$ the set-valued mapping $\mathcal{X}_\alpha : \Omega \rightarrow \mathcal{K}_c(\mathbb{R}) = \text{space of nonempty compact intervals}$, with $\mathcal{X}_\alpha(\omega) = (\mathcal{X}(\omega))_\alpha$, is a compact random interval.

In the fuzzy number case, a random fuzzy number associated with $(\Omega, \mathcal{A}, P)$ can be alternatively defined as a mapping $\mathcal{X} : \Omega \rightarrow \mathcal{F}_c^\circ(\mathbb{R})$ such that

- either $s_{\mathcal{X}} : \Omega \rightarrow \mathcal{L}([-1, 1] \times (0, 1])$ is an $\mathcal{L}([-1, 1] \times (0, 1])$ -valued random element (that is, $s_{\mathcal{X}}$ is measurable with respect to the Borel σ -field generated by the topology induced by the metric associated with $\|\cdot\|_\theta^\phi$ on $\mathcal{L}([-1, 1] \times (0, 1])$);
- or for all $\alpha \in (0, 1]$ and $u \in [-1, 1]$, the real-valued function $s_{\mathcal{X}}(u, \alpha) : \Omega \rightarrow \mathbb{R}$ is a (real-valued) random variable;
- or for all $\alpha \in (0, 1]$ the real-valued functions $\inf \mathcal{X}_\alpha : \Omega \rightarrow \mathbb{R}$ and $\sup \mathcal{X}_\alpha : \Omega \rightarrow \mathbb{R}$ are (real-valued) random variables such that $\inf \mathcal{X}_\alpha \leq \sup \mathcal{X}_\alpha$;
- or for all $\alpha \in (0, 1]$ the real-valued functions $\text{mid } \mathcal{X}_\alpha : \Omega \rightarrow \mathbb{R}$ and $\text{spr } \mathcal{X}_\alpha : \Omega \rightarrow [0, +\infty)$ are (real-valued) random variables.

Random fuzzy numbers are properly employed when there exists a random mechanism which generates fuzzy number data, that is, when the random attribute we deal with is intrinsically fuzzy number-valued. In other words, the ‘ontic’ view is taken, so that the involved fuzziness is supposed to be essentially associated with the data. For instance, in rating the interest of a given course, the taste of a dessert, and so on, fuzzy numbers offer a suitable and versatile tool to express the interest, taste, etc. which are not existing real-valued variables. This is often made by considering a few linguistic terms, which are later identified by means of fuzzy numbers (see, for instance, Herrera et al. [31]). Actually, one can draw fuzzy numbers capturing better the ratings in an unconstrained way.

On the other hand, in rating the price of the main courses in the menu of a restaurant, one can respond by using linguistic terms. This corresponds to what is called ‘epistemic’ view; there exists an underlying real-valued random variable, which is the price of the main course in the menus of different restaurants.

For a recent and clarifying discussion on the epistemic and ontic fuzzy sets, readers are encouraged to look at Dubois and Prade [13]. Most of the real-life examples one can think in connection with random fuzzy numbers involve human valuations which frequently arise in Social Sciences, Medicine, Decision Making, Control Engineering, and so on.

Formally, the approach could also be applied to some situations related to the epistemic view (more concretely, when fuzzy data are assumed to come from a vague perception or report of an exact measurement or observation in an experimental performance). However, other suitable approaches have been proposed in the literature to specifically address these situations (see, for instance, Couso and Sánchez [7], Denœux et al. [9], and Denœux [8]).

Moreover, one should be aware that the statistical analysis with Féron/Puri and Ralescu’s model should be focused on drawing conclusions on the random fuzzy numbers generating the fuzzy data. For instance, when fuzzy data correspond to fuzzy perceptions of values of a one-dimensional random variable involved in a random experiment, the concept of fuzzy random variable in Kruse and Meyer’s sense [37] and the related statistical methodology are mostly focused on the distribution of the random experiment producing real-valued data (the so-called – unknown – original random variable). In this respect, it should be pointed out that if, when dealing with fuzzy data corresponding to fuzzy perceptions of values of a real-valued random variable, we wish to consider Féron/Puri and Ralescu’s approach and the associated statistical methods, conclusions could be ‘technically’ drawn; however, these conclusions would only refer to the distribution of the random mechanism producing the fuzzy data instead of to the original real-valued one.

Of course, there are many other situations in which the Féron/Puri and Ralescu’s approach does not make sense at all (for instance, when a possibility distribution representing what is known about some data cannot be considered to have been generated at random; see the detailed discussion of this point in Denœux [8]).

One can clearly assert that when Féron/Puri and Ralescu’s approach is used, the distribution of the random fuzzy numbers can be systematically induced on the basis of the Borel-measurability assumption, so there is no need to introduce a new concept for this purpose. Furthermore, related notions like the stochastic independence of random fuzzy sets, the equality of distributions (and hence, the concept of a simple random sample from a random fuzzy number, which will be considered in the next section) will be immediately induced.

Random fuzzy numbers extend the existing theories of random variables and random intervals, and they allow us to develop statistical analysis of fuzzy number data within an appropriate probabilistic setting and to preserve the key ideas and tools from the real-valued case (e.g., unbiasedness, mean squared error, consistency, etc. associated with a fuzzy-valued estimator, or the probabilities of Type I and Type II errors, p -values, etc., associated with testing statistical hypotheses).

In analyzing fuzzy data, especially from an inferential viewpoint, the two most relevant characteristics (summary measures) associated with the distribution of a random fuzzy number are the mean and the variance. They will be assumed to be based on Fréchet’s generalization for abstract metric spaces [18], so that both are related to a great or a full extent to D_θ^ϕ . In this way, we will assume the Aumann-type mean value to be defined as the fuzzy number minimizing a convex combination of the squared D_θ^ϕ -distances to the random fuzzy number values (a mean squared distance or error measure), and the variance will be considered to be the corresponding minimum.

More concretely, the Aumann-type mean value of a random fuzzy number can be presented in several equivalent ways, each of them being associated with one of the considered definitions for random fuzzy numbers.

Definition 3.2. Given a probability space $(\Omega, \mathcal{A}, P)$ and $\mathcal{X} : \Omega \rightarrow \mathcal{F}_c^\circ(\mathbb{R})$ an associated random fuzzy number, the Aumann-type mean value of $\mathcal{X}$ is the fuzzy number $\tilde{E}(\mathcal{X}) \in \mathcal{F}_c^\circ(\mathbb{R})$, if it exists, such that

$$\tilde{E}(\mathcal{X}) = \arg \min_{\tilde{U} \in \mathcal{F}_c^\circ(\mathbb{R})} E([D_\theta^\varphi(\mathcal{X}, \tilde{U})]^2).$$

Equivalently, for all $\alpha \in (0, 1]$

$$(\tilde{E}(\mathcal{X}))_\alpha = \{E(X) : X : \Omega \rightarrow \mathbb{R}, X \in L^1(\Omega, \mathcal{A}, P), X \in \mathcal{X}_\alpha \text{ a.s. } [P]\} \quad (7)$$

with $L^1(\Omega, \mathcal{A}, P)$ being the class of the integrable functions w.r.t. $(\Omega, \mathcal{A}, P)$.

Let $\mathcal{X} : \Omega \rightarrow \mathcal{F}_c^\circ(\mathbb{R})$ be a random fuzzy number. If the Aumann-type mean value in Eq. (7) exists, it can equivalently be defined as the fuzzy number satisfying any of the following connections:

- $s_{\tilde{E}(\mathcal{X})} = E(s_{\mathcal{X}})$;
- $\inf(\tilde{E}(\mathcal{X}))_\alpha = E(\inf \mathcal{X}_\alpha)$ and $\sup(\tilde{E}(\mathcal{X}))_\alpha = E(\sup \mathcal{X}_\alpha)$;
- $\text{mid}(\tilde{E}(\mathcal{X}))_\alpha = E(\text{mid} \mathcal{X}_\alpha)$ and $\text{spr}(\tilde{E}(\mathcal{X}))_\alpha = E(\text{spr} \mathcal{X}_\alpha)$.

The Aumann-type mean value of a random fuzzy number is well-defined when for all $\alpha \in (0, 1]$ the real-valued functions $\inf \mathcal{X}_\alpha, \sup \mathcal{X}_\alpha$ belong to $L^1(\Omega, \mathcal{A}, P)$ (i.e., they are integrable functions in $(\Omega, \mathcal{A}, P)$). Furthermore, it is equivariant under ‘linear’ (in accordance with the usual fuzzy arithmetic) transformations, it is additive and it is coherent with the usual arithmetic with fuzzy numbers. It is also supported by Strong Laws of Large Numbers.

On the other hand, the variance of a random fuzzy number in accordance with Fréchet’s approach can be quantified (see Lubiano et al. [40]) by the minimum $\min_{\tilde{U} \in \mathcal{F}_c^\circ(\mathbb{R})} E([D_\theta^\varphi(\mathcal{X}, \tilde{U})]^2)$, and it is a real-valued measure of the mean ‘error’ incurred in approximating/representing the random fuzzy number values by means of the corresponding Aumann-type mean value. Thus,

Definition 3.3. Given a probability space $(\Omega, \mathcal{A}, P)$ and a random fuzzy number $\mathcal{X}$ for which $\tilde{E}(\mathcal{X}) \in \mathcal{F}_c^\circ(\mathbb{R})$ exists, the (θ, φ) -Fréchet variance of $\mathcal{X}$ is the real number, if it exists, given by

$$\sigma_{\mathcal{X}}^2 = E([D_\theta^\varphi(\mathcal{X}, \tilde{E}(\mathcal{X}))]^2). \quad (8)$$

The (θ, φ) -Fréchet variance of a random fuzzy number in Eq. (8) can be equivalently defined as the real number, if it exists, such that

$$\sigma_{\mathcal{X}}^2 = \text{Var}(s_{\mathcal{X}}) = \int_{(0,1]} (\text{Var}(\text{mid} \mathcal{X}_\alpha) + \theta \cdot \text{Var}(\text{spr} \mathcal{X}_\alpha)) d\varphi(\alpha).$$

The (θ, φ) -Fréchet variance vanishes only for degenerate random fuzzy numbers (i.e., random fuzzy numbers being almost surely equal to a given fuzzy number), it is translational and rotational invariant and it is additive in dealing with independent random fuzzy numbers (i.e., if $\mathcal{X}$ and $\mathcal{Y}$ are independent random fuzzy numbers associated with the same probability space $(\Omega, \mathcal{A}, P)$ and such that $\sigma_{\mathcal{X}}^2$ and $\sigma_{\mathcal{Y}}^2$ exist, we have that $\sigma_{\mathcal{X}+\mathcal{Y}}^2 = \sigma_{\mathcal{X}}^2 + \sigma_{\mathcal{Y}}^2$).

Another summary measure associated with the distribution of random fuzzy numbers is the covariance, which cannot be defined directly on the basis of D_θ^φ , but it can be formalized by using the ideas in Näther [43], Körner and Näther [36], González-Rodríguez et al. [23] and Blanco-Fernández et al. [2], which are based on the aforementioned isometrical embedding. Since the space $\mathcal{L}(\{-1, 1\} \times (0, 1])$ is a Hilbert one, it has a linear structure, and hence the covariance of random fuzzy numbers can be defined by involving external operations like the inner product $\langle \cdot, \cdot \rangle_\theta^\varphi$ on $\mathcal{L}(\{-1, 1\} \times (0, 1])$ given by

$$\begin{aligned} \langle f, g \rangle_\theta^\varphi = & \int_{(0,1]} \left(\left[\frac{f(1, \alpha) - f(-1, \alpha)}{2} \right] \cdot \left[\frac{g(1, \alpha) - g(-1, \alpha)}{2} \right] \right. \\ & \left. + \theta \left[\frac{f(1, \alpha) + f(-1, \alpha)}{2} \right] \cdot \left[\frac{g(1, \alpha) + g(-1, \alpha)}{2} \right] \right) d\varphi(\alpha) \end{aligned}$$

(i.e., $\|f\|_\theta^\varphi = \sqrt{\langle f, f \rangle_\theta^\varphi}$). In this way,

Definition 3.4. If $\mathcal{X}, \mathcal{Y}$ are two random fuzzy numbers, the (θ, φ) -covariance between $\mathcal{X}$ and $\mathcal{Y}$ is defined, if it exists, as the real value

$$\sigma_{\mathcal{X}, \mathcal{Y}} = E(\langle s_{\mathcal{X}} - s_{E(\mathcal{X})}, s_{\mathcal{Y}} - s_{E(\mathcal{Y})} \rangle_{\theta}^{\varphi}) = E(\langle s_{\mathcal{X}} - E(s_{\mathcal{X}}), s_{\mathcal{Y}} - E(s_{\mathcal{Y}}) \rangle_{\theta}^{\varphi}).$$

Equivalently, if $\sigma_{\mathcal{X}, \mathcal{Y}}$ exists

$$\sigma_{\mathcal{X}, \mathcal{Y}} = \text{Cov}(s_{\mathcal{X}}, s_{\mathcal{Y}}) = \int_{(0,1]} (\text{Cov}(\text{mid } \mathcal{X}_{\alpha}, \text{mid } \mathcal{Y}_{\alpha}) + \theta \cdot \text{Cov}(\text{spr } \mathcal{X}_{\alpha}, \text{spr } \mathcal{Y}_{\alpha})) d\varphi(\alpha). \quad (9)$$

The covariance of two random fuzzy numbers in Eq. (9) preserves several properties from the real-valued case. Thus, it vanishes in case of independent random fuzzy sets. However, in general $\sigma_{\mathcal{Y}, \mathcal{X}} \neq \gamma \sigma_{\mathcal{X}, \mathcal{Y}}$.

The covariance of two random fuzzy numbers is basically introduced as a valuable tool in the development of regression studies for linear models with random fuzzy numbers, where the linearity is formalized in terms of the usual fuzzy arithmetic. More concretely, as it will be later seen in Theorem 4.3, the covariance is involved in formalizing the optimal solution of the regression coefficient.

4. Some statistical methods to analyze fuzzy data on the basis of the (θ, φ) -distance

As we have already commented, in developing statistical analysis methods to deal with fuzzy number data several difficulties arise, namely, those associated with the lack of a well-defined difference between fuzzy numbers which preserves the properties of the real-valued case, the lack of realistic and widely applicable distribution models for random fuzzy numbers, the lack of general asymptotic results to be directly applied for inferential purposes, etc.

At first glance one might be tempted to consider the following reasoning: in extending statistical methods dealing with real-valued data to either interval-valued or fuzzy number-valued data one can take into account two points:

- Any interval value can be identified with a certain two-dimensional vector (by considering either its infimum/supremum or its mid-point/spread representation);
- Any fuzzy number can be identified with its real-valued support function,

so we can immediately particularize the well-known multivariate and functional procedures, respectively, to analyze these data.

Unfortunately, the last reasoning fails due to the nonlinearities of the space of interval values and the space of fuzzy numbers. Multivariate and Functional Data Analysis techniques take advantage of the Hilbert structure of the spaces in which they are defined ($\mathbb{R}^2$ and $\mathcal{L}([-1, 1] \times (0, 1])$ in the interval and fuzzy number identification, respectively). However, any of the vectorial representations of interval values (e.g., the $K \mapsto (\text{mid } K, \text{spr } K)$) or the functional representation of fuzzy number values through the support function, $\tilde{U} \mapsto s_{\tilde{U}}$, embed isometrically $\mathcal{K}_c(\mathbb{R})$ and $\mathcal{F}_c^{\circ}(\mathbb{R})$ into closed convex cones of Hilbert spaces, which are not Hilbert spaces. In other words, the vectors and functions associated with interval and fuzzy values through the isometries should fulfill some constraints, since not all the elements either in $\mathbb{R}^2$ or in $\mathcal{L}([-1, 1] \times (0, 1])$ determine a compact interval or a fuzzy number.

Consequently, if one tries to particularize directly well-known multivariate and functional statistical procedures to analyze interval or fuzzy data, respectively, one should never forget that the mathematical and probabilistic results and developments supporting these procedures neglect such constraints. So, in general the particularization is wrong (irrespective of the fact that some procedures could be ‘physically’ applied, but conclusions from them would be neither reliable nor well-supported).

In summary, when dealing with interval or fuzzy number data, one should not try to particularize directly any available method to analyze multivariate or functional data. The suitable way to proceed in developing inferential methods will be

- (i) Either designing *ad hoc* procedures like, for instance,
 - when dealing with simple (i.e., taking a finite number of different values) random fuzzy numbers, by applying Large Sample Theory to the vector parameter characterizing its probability distribution, and approximating later the asymptotic results by means of bootstrap techniques, like in Montenegro et al. [42],
 - when dealing with general random fuzzy numbers, by applying the bootstrapped Central Limit Theorem for general space-valued random elements, like in González-Rodríguez et al. [30];
- (ii) Or designing and developing procedures
 - for multivariate random vectors under restrictions, these restrictions including those required for the interval, set and fuzzy set values, so that they can be particularized to deal with this type of data, or
 - for Hilbert space-valued random elements (which are based on appropriate tools, like the bootstrapped Central Limit Theorem by Giné and Zinn [22]) ensuring that once they are particularized to the fuzzy number case the involved random elements fall within the class of random fuzzy numbers, like in González-Rodríguez et al. [26].

Unfortunately, at present only a very few of these last procedures have been stated.

In this section we will summarize three main problems concerning the analysis of fuzzy data, as well as some existing solutions developed along the last years and being unequivocally related to D_θ^φ .

4.1. A (θ, φ) -distance-based testing about the mean values of random fuzzy numbers

When testing the null hypothesis that either the population mean of a random fuzzy number equals a given fuzzy number (one-sample case), or the population means of two (independent or linked) random fuzzy numbers coincide (two-sample case), or the equality of the population means of k (independent or linked) random fuzzy numbers holds (ANOVA case), the null hypothesis can be equivalently expressed by stating that either the D_θ^φ -distance between the mean and the hypothetical value equals 0, or the D_θ^φ -distance between the means equals 0, or the sum of the squared D_θ^φ -distances between the means and the ‘overall mean’ equals 0, respectively. These equivalences inspire methods to test the original null hypotheses.

Some of the relevant papers related to testing about the means of random fuzzy numbers (most of them related to the more general case of random fuzzy sets) are Körner [35], Montenegro et al. [42] and González-Rodríguez et al. [30] for the one-sample case, Montenegro et al. [41] and González-Rodríguez et al. [28] for the two-sample case, and Gil et al. [21], González-Rodríguez et al. [26], for the k -sample case.

To illustrate one of the concerned methods, we are now recalling the one established in González-Rodríguez et al. [26] (the k -sample case), which has followed the reasoning scheme described in (ii) above. More precisely, we will recall first the supporting result for Hilbert space-valued random elements and particularize it to give an algorithm for the ANOVA test with independent random fuzzy numbers.

Theorem 4.1. Let $\mathcal{H}$ be a separable Hilbert space with inner product $\langle \cdot, \cdot \rangle$ and norm $\| \cdot \|$. Let $H_1, \dots, H_k$ be independent $\mathcal{H}$ -valued random elements with means $m_1, \dots, m_k$ and covariance functions $K_1, \dots, K_k$, respectively. Let $\{H_{ij}\}_{j=1}^{n_i}$ be a simple random sample obtained from H_i for all $i \in \{1, \dots, k\}$ and let $n = n_1 + \dots + n_k$.

If $m_1 = \dots = m_k$, and $n_i \rightarrow \infty$ and $\frac{n_i}{n} \rightarrow p_i > 0$ as $n \rightarrow \infty$ for all $i \in \{1, \dots, k\}$, then as $n \rightarrow \infty$

- the natural extension of the ANOVA test statistic satisfies that

$$A_n = \sum_{i=1}^k n_i \|\bar{H}_i - \bar{H}_{..}\|^2 \xrightarrow{\mathcal{L}} \sum_{i=1}^k \left\| Z_i - \sum_{l=1}^k \alpha_{li} Z_l \right\|^2 = A,$$

where $\bar{H}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} H_{ij}$, $\bar{H}_{..} = \frac{1}{n} \sum_{i=1}^k \sum_{j=1}^{n_i} H_{ij}$, $\alpha_{li} = \sqrt{p_l/p_i} / \sum_{r=1}^k (p_r/p_i)$, and $Z_1, \dots, Z_k$ are centered Gaussian processes over $\mathcal{H}$ with covariance functions $K_1, \dots, K_k$;

- if further H_i is non-degenerated for some i ,

$$B_n = \frac{1}{n} \sum_{i=1}^k \sum_{j=1}^{n_i} \|H_{ij} - \bar{H}_i\|^2 \xrightarrow{\text{a.s. } [P]} \sum_{i=1}^k p_i E \|H_i - m_i\|^2 > 0;$$

- whence the Studentized statistic

$$T_n = \frac{\sum_{i=1}^k n_i \|\bar{H}_i - \bar{H}_{..}\|^2}{\frac{1}{n} \sum_{i=1}^k \sum_{j=1}^{n_i} \|H_{ij} - \bar{H}_i\|^2} \xrightarrow{\mathcal{L}} \frac{\sum_{i=1}^k \|Z_i - \sum_{l=1}^k \alpha_{li} Z_l\|^2}{\sum_{i=1}^k p_i E \|H_i - m_i\|^2} = T.$$

On the other hand, let $\{H_{ij}^*\}_{j=1}^{n_i}$ be a simple random sample drawn at random from $\{H_{i1} - \bar{H}_i, \dots, H_{in_i} - \bar{H}_i\}$ for each $i \in \{1, \dots, k\}$. Then, $\sqrt{n_i} \cdot \bar{H}_i^*$ converges in law to Z_i as $n_i \rightarrow \infty$, and if $m_1 = \dots = m_k$, and $n_i \rightarrow \infty$ and $\frac{n_i}{n} \rightarrow p_i > 0$ as $n \rightarrow \infty$ for all $i \in \{1, \dots, k\}$, then as $n \rightarrow \infty$ the bootstrap statistic

$$T_n^* = \frac{\sum_{i=1}^k n_i \|\bar{H}_i^* - \bar{H}_{..}^*\|^2}{\frac{1}{n} \sum_{i=1}^k \sum_{j=1}^{n_i} \|H_{ij}^* - \bar{H}_i^*\|^2} \xrightarrow{\mathcal{L}} T \quad \text{a.s. } [P].$$

The preceding result can be particularized to random fuzzy numbers to get Algorithm 1 to test the null hypothesis

$$H_0: \tilde{E}(\mathcal{X}_1) = \dots = \tilde{E}(\mathcal{X}_k),$$

on the basis of samples of independent observations $(\tilde{x}_{i1}, \dots, \tilde{x}_{in_i})$ from the random fuzzy numbers $\mathcal{X}_i$, $i \in \{1, \dots, k\}$.

In González-Rodríguez et al. [26], the consistency of the test above under local alternatives has been proved. It should also be pointed out that the involved crossed sums of means in the numerator have been considered to ensure, in accordance with the bootstrapped Central Limit Theorem by Giné and Zinn [22], that the null hypothesis is fulfilled.

Step 1.

Compute the statistic value:

$$\mathcal{T}_n = \frac{\sum_{i=1}^k n_i [D_\theta^\varphi(\bar{x}_i, \bar{x}_{..})]^2}{\sum_{i=1}^k \frac{1}{n_i^2} \sum_{j=1}^{n_i} [D_\theta^\varphi(\tilde{x}_{ij}, \bar{x}_{i.})]^2}, \quad \bar{x}_{i.} = \frac{1}{n_i} \sum_{j=1}^{n_i} \tilde{x}_{ij}, \quad \bar{x}_{..} = \frac{1}{n} \sum_{i=1}^k \sum_{j=1}^{n_i} \tilde{x}_{ij}.$$

Step 2.

Obtain a sample of independent observations $(\tilde{x}_{i1}^*, \dots, \tilde{x}_{in_i}^*)$ from $\{\tilde{x}_{i1}, \dots, \tilde{x}_{in_i}\}$.

Step 3.

Compute the bootstrap statistic value

$$\mathcal{T}_n^* = \frac{\sum_{i=1}^k n_i [D_\theta^\varphi(\bar{x}_{i.}^* + \bar{x}_{..}, \bar{x}_{i.} + \bar{x}_{..})]^2}{\sum_{i=1}^k \frac{1}{n_i^2} \sum_{j=1}^{n_i} [D_\theta^\varphi(\tilde{x}_{ij}^*, \bar{x}_{i.}^*)]^2}.$$

Step 4.

Repeat Steps 2 and 3 a large number of times B to obtain B values $\{(\mathcal{T}_n^*)_1, \dots, (\mathcal{T}_n^*)_B\}$.

Step 5.

Approximate the p -value of the test as the proportion of values in $\{(\mathcal{T}_n^*)_1, \dots, (\mathcal{T}_n^*)_B\}$ which are greater than or equal to $\mathcal{T}_n$.

Algorithm 1. Bootstrapped ANOVA test algorithm for random fuzzy numbers.

As we have already pointed out, if we simply try to particularize the asymptotic results in the ANOVA test to the case of fuzzy data (more precisely, to support functions of fuzzy data) the limit process in the asymptotic result is not $\mathcal{F}_c^\circ(\mathbb{R})$ -valued. However, the suggested bootstrap approximation can be expressed in terms of sums and distances, whence the bootstrap statistic is defined within the sampling cone. The results in the ANOVA test could be applied to cluster random fuzzy numbers (see González-Rodríguez et al. [25], Colubi et al. [6]).

4.2. Testing about the (θ, φ) -Fréchet variances of random fuzzy numbers

When testing the null hypothesis that either the (θ, φ) -Fréchet variance of a random fuzzy number equals a given non-negative number (one-sample case), or the (θ, φ) -Fréchet variances of two (independent or linked) random fuzzy numbers coincide (two-sample case), or the equality of the (θ, φ) -Fréchet variances of k (independent or linked) random fuzzy numbers holds (homoscedasticity hypothesis), the null hypothesis refers to the equality of real numbers. Prompted by either the theory started by Levene [38] and Draper and Hunter [11] on the homoscedasticity testing or the U -statistics on Hilbert spaces by Hoeffding [32], some results have been obtained.

Some of the relevant papers related to testing about the (θ, φ) -Fréchet variances of a random fuzzy numbers (most of them related to the more general case of random fuzzy sets) are Lubiano et al. [39] and Ramos-Guajardo et al. [47] for the one-sample case, and Ramos-Guajardo and Lubiano [49] for the two and k -sample cases.

To illustrate one of the concerned methods, we are now recalling the one established in Ramos-Guajardo and Lubiano [49] (the k -sample case), which has followed the reasoning scheme described in (i) above.

Theorem 4.2. Let $\mathcal{X}_1, \dots, \mathcal{X}_k$ be independent random fuzzy numbers with (θ, φ) -Fréchet variances $\sigma_{\mathcal{X}_1}^2, \dots, \sigma_{\mathcal{X}_k}^2$, respectively, and let $\{\mathcal{X}_{ij}\}_{j=1}^{n_i}$ be a simple random sample obtained from $\mathcal{X}_i$ for all $i \in \{1, \dots, k\}$, where $n = n_1 + \dots + n_k$.

If $\sigma_{\mathcal{X}_1}^2 = \dots = \sigma_{\mathcal{X}_k}^2$, $E(\|\mathcal{X}_i\|_\theta^\varphi)^4 < \infty$ and $\frac{n_i}{n} \rightarrow p_i \in (0, 1)$ as $n_i \rightarrow \infty$ for all $i \in \{1, \dots, k\}$, then as $n_i \rightarrow \infty$ for all i we have that

$$\begin{aligned} T_{(n_1, \dots, n_k)} &= \frac{\sum_{i=1}^k n_i (\widehat{\sigma_{\mathcal{X}_i}^2} - \widehat{\sigma^2})^2}{\sum_{i=1}^k \frac{1}{n_i} \sum_{j=1}^{n_i} ([D_\theta^\varphi(\mathcal{X}_{ij}, \bar{\mathcal{X}}_{i.})]^2 - \frac{1}{n_i} \sum_{j=1}^{n_i} [D_\theta^\varphi(\mathcal{X}_{ij}, \bar{\mathcal{X}}_{i.})]^2)^2} \\ &\xrightarrow{\mathcal{L}} \frac{\sum_{i=1}^k (Y_i - \sum_{l=1}^k \sqrt{p_l p_i} Y_l)^2}{\sum_{i=1}^k \frac{1}{n_i} \sum_{j=1}^{n_i} ([D_\theta^\varphi(\mathcal{X}_{ij}, \tilde{E}(\mathcal{X}_i))]^2 - \frac{1}{n_i} \sum_{j=1}^{n_i} [D_\theta^\varphi(\mathcal{X}_{ij}, \tilde{E}(\mathcal{X}_i))]^2)^2} = T, \end{aligned}$$

with

$$\widehat{\sigma_{\mathcal{X}_i}^2} = \frac{1}{n_i} \sum_{j=1}^{n_i} [D_\theta^\varphi(\mathcal{X}_{ij}, \bar{\mathcal{X}}_{i.})]^2, \quad \widehat{\sigma^2} = \frac{1}{n} \sum_{i=1}^k \sum_{j=1}^{n_i} [D_\theta^\varphi(\mathcal{X}_{ij}, \bar{\mathcal{X}}_{i.})]^2, \quad \bar{\mathcal{X}}_{i.} = \frac{1}{n_i} \sum_{j=1}^{n_i} \mathcal{X}_{ij},$$

and the random vector $(Y_1, \dots, Y_k)^t$ being normally distributed as an $\mathcal{N}_k(\mathbf{0}, \Sigma)$, with covariance matrix

$$\Sigma = \text{diag} \left(\frac{1}{n_1} \sum_{j=1}^{n_1} \left([D_{\theta}^{\varphi}(\mathcal{X}_{1j}, \tilde{E}(\mathcal{X}_1))]^2 - \frac{1}{n_1} \sum_{j=1}^{n_1} [D_{\theta}^{\varphi}(\mathcal{X}_{1j}, \tilde{E}(\mathcal{X}_1))]^2 \right)^2, \right. \\ \left. \dots, \frac{1}{n_k} \sum_{j=1}^{n_k} \left([D_{\theta}^{\varphi}(\mathcal{X}_{kj}, \tilde{E}(\mathcal{X}_k))]^2 - \frac{1}{n_k} \sum_{j=1}^{n_k} [D_{\theta}^{\varphi}(\mathcal{X}_{kj}, \tilde{E}(\mathcal{X}_k))]^2 \right)^2 \right).$$

On the other hand, let $\{\mathcal{X}_{ij}^*\}_{j=1}^{n_i}$ be a simple random sample drawn at random from $\{\mathcal{X}_{i1}, \dots, \mathcal{X}_{in_i}\}$ for each $i \in \{1, \dots, k\}$. Then, the null hypothesis

$$H_0: \sigma_{\mathcal{X}_1}^2 = \dots = \sigma_{\mathcal{X}_k}^2$$

must be rejected at (approximately) the significance level α whenever $T_{(n_1, \dots, n_k)} > t_{(1-\alpha)}^*$ where $t_{(1-\alpha)}^*$ is the $(1 - \alpha)$ -quantile of the distribution of the bootstrap statistic

$$T_{(n_1, \dots, n_k)}^* = \frac{\sum_{i=1}^k n_i \left[\frac{1}{\sigma_{\mathcal{X}_i}^2} (\widehat{\sigma_{\mathcal{X}_i}^2} - \widehat{\sigma_{\mathcal{X}_i}^{*2}}) - \frac{1}{n} \sum_{l=1}^k n_l \left(\frac{1}{\sigma_{\mathcal{X}_l}^2} (\widehat{\sigma_{\mathcal{X}_l}^2} - \widehat{\sigma_{\mathcal{X}_l}^{*2}}) \right) \right]^2}{\sum_{i=1}^k \frac{1}{n_i \sigma_{\mathcal{X}_i}^4} \sum_{j=1}^{n_i} [(D_{\theta}^{\varphi}(\mathcal{X}_{ij}^*, \overline{\mathcal{X}_i^*}))^2 - \widehat{\sigma_{\mathcal{X}_i}^{*2}}]^2},$$

the distribution of $T_{(n_1, \dots, n_k)}^*$ being approximated by Monte Carlo method.

The testing procedure based on the preceding result could be presented in an algorithmic way as for the one in testing the equality of means.

The capability of the test is analyzed by using a sequence of alternatives converging to the null hypothesis as the sample size increases (i.e., by considering local alternatives) by Ramos-Guajardo et al. [48].

4.3. A (θ, φ) -distance-based least-squares approach to the simple linear regression analysis with random fuzzy numbers

When considering the regression analysis with fuzzy number data, the least-squares method based on the (θ, φ) -distance has been examined in case of relationships like the arithmetic-based linear one. More concretely, let $(\mathcal{X}, \mathcal{Y})$ be a two-dimensional random vector of fuzzy numbers (i.e., $\mathcal{X}$ and $\mathcal{Y}$ are two random fuzzy numbers) associated with a probability space $(\Omega, \mathcal{A}, P)$, $\mathcal{X}$ and $\mathcal{Y}$ having finite positive (θ, φ) -Fréchet's variances. The simple linear model to be considered will be

$$\mathcal{Y} = a \cdot \mathcal{X} + \mathcal{E},$$

where $a \in \mathbb{R}$ and $\mathcal{E}$ is a random fuzzy number with finite variance and $\tilde{E}(\mathcal{E}|\mathcal{X}) = \tilde{B} \in \mathcal{F}_c^{\circ}(\mathbb{R})$.

The population regression function corresponds to the conditional expectation

$$\tilde{E}(\mathcal{Y}|\mathcal{X}) = a \cdot \mathcal{X} + \tilde{B}.$$

Arguments supporting the independent term in the linear model to be included in the formalization of the possible errors, as well as the fact that the linear relationship between two random fuzzy numbers can be expressed as a function depending on at most two different parameters 'a', can be found in González-Rodríguez et al. [23].

The (θ, φ) -distance-based least-squares approach to this linear regression problem first searches for

$$(\hat{a}, \hat{\tilde{B}}) = \arg \min_{(a, \tilde{B}) \in \mathbb{R} \times \mathcal{F}_c(\mathbb{R})} \frac{1}{n} \sum_{i=1}^n [D_{\theta}^{\varphi}(\mathcal{Y}_i, a \cdot \mathcal{X}_i + \tilde{B})]^2,$$

where $((\mathcal{X}_1, \mathcal{Y}_1), \dots, (\mathcal{X}_n, \mathcal{Y}_n))$ is a simple random sample from $(\mathcal{X}, \mathcal{Y})$. That is, the first aim is to find the least-squares estimators (in the sense of the considered metric) of the unknown parameters a and $\tilde{B}$ on the basis of the simple random sample and under the assumption that the linear model is satisfied, i.e., $\mathcal{Y}_i = a \cdot \mathcal{X}_i + \mathcal{E}_i$ for all $i \in \{1, \dots, n\}$.

A second aim is testing hypothesis like the linearity of the relationship (i.e., the null hypothesis of a in the linear model vanishing).

Furthermore, the definition of a linear determination coefficient and other linear models allowing for a higher flexibility have also been pointed out in the literature.

Studies about these problems have been developed in the particular case of random intervals, but the fuzzy number case is still open, especially in connection with the last three questions. Also multiple regression has been recently analyzed in case of real-valued explanatory variables and fuzzy-valued response (see, for instance, Ferraro et al. [17]).

To illustrate some of the conclusions about the least-squares estimators in the linear model $\mathcal{Y} = a \cdot \mathcal{X} + \mathcal{E}$ we can consider the following:

Theorem 4.3. Let $\mathcal{X}$ and $\mathcal{Y}$ be two random fuzzy numbers associated with a probability space $(\Omega, \mathcal{A}, P)$ with finite positive (θ, φ) -Fréchet's variances. Consider the linear model $\mathcal{Y} = a \cdot \mathcal{X} + \mathcal{E}$, with $a \in \mathbb{R}$ and $\mathcal{E}$ being a random fuzzy number with finite variance and $\tilde{E}(\mathcal{E}|\mathcal{X}) = \tilde{B} \in \mathcal{F}_c^\circ(\mathbb{R})$. On the basis of a simple random sample $\{(\mathcal{X}_i, \mathcal{Y}_i)\}_{i=1}^n$ obtained from $(\mathcal{X}, \mathcal{Y})$, the least-squares estimators of a and $\tilde{B}$ are given by

$$\hat{a} = \begin{cases} \frac{v \cdot \sum_{i=1}^n \text{Cov}(\mathcal{X}_i, \mathcal{Y}_i) - \beta \cdot \sum_{i=1}^n \text{Cov}(-\mathcal{X}_i, \mathcal{Y}_i)}{\sum_{i=1}^n [D_\theta^\varphi(\mathcal{X}_i, \bar{\mathcal{X}})]^2} & \text{if } v \cdot \beta = 0, \\ \left. \begin{aligned} & -\beta \cdot \frac{\sum_{i=1}^n \text{Cov}(-\mathcal{X}_i, \mathcal{Y}_i)}{\sum_{i=1}^n [D_\theta^\varphi(\mathcal{X}_i, \bar{\mathcal{X}})]^2} & \text{if } \frac{\sum_{i=1}^n \text{Cov}(-\mathcal{X}_i, \mathcal{Y}_i)}{\sum_{i=1}^n \text{Cov}(\mathcal{X}_i, \mathcal{Y}_i)} \geq \frac{2v - v^2}{2\beta - \beta^2} \\ & v \cdot \frac{\sum_{i=1}^n \text{Cov}(\mathcal{X}_i, \mathcal{Y}_i)}{\sum_{i=1}^n [D_\theta^\varphi(\mathcal{X}_i, \bar{\mathcal{X}})]^2} & \text{otherwise} \end{aligned} \right\} & \text{if } v \cdot \beta \neq 0 \end{cases}$$

and $\tilde{B}$ such that for each $\alpha \in (0, 1]$ we have $\tilde{B}_\alpha = [\inf \mathcal{Y}_\alpha - \hat{a} \cdot \inf \mathcal{X}_\alpha, \sup \mathcal{Y}_\alpha - \hat{a} \cdot \sup \mathcal{X}_\alpha]$, where

$$v = \begin{cases} \min\{1, \frac{a_0 \sum_{i=1}^n [D_\theta^\varphi(\mathcal{X}_i, \bar{\mathcal{X}})]^2}{\sum_{i=1}^n \text{Cov}(-\mathcal{X}_i, \mathcal{Y}_i)}\} & \text{if } \sum_{i=1}^n \text{Cov}(-\mathcal{X}_i, \mathcal{Y}_i) > 0, \\ 0 & \text{otherwise,} \end{cases}$$

$$\beta = \begin{cases} \min\{1, \frac{a^0 \sum_{i=1}^n [D_\theta^\varphi(\mathcal{X}_i, \bar{\mathcal{X}})]^2}{\sum_{i=1}^n \text{Cov}(\mathcal{X}_i, \mathcal{Y}_i)}\} & \text{if } \sum_{i=1}^n \text{Cov}(\mathcal{X}_i, \mathcal{Y}_i) > 0, \\ 0 & \text{otherwise,} \end{cases}$$

$[-a_0, a^0]$ being the feasible set of solutions for the parameter a under the assumed linearity relationship, whenever not all the $\mathcal{X}_i$ reduce simultaneously to real numbers.

Of course, this section did not aim at covering all the relevant literature and methods on the topic, but we have simply considered a representative sample of them.

It should be remarked that, in addition to providing us with tools to draw statistical conclusions when available data are intrinsically fuzzy number-valued, this methodology can also be considered to draw some conclusions when available data are real-valued and exactly known. In this way, a relevant application of the distance D_θ^φ is that of stating a statistical distance between two real-valued random variables (i.e., a measure to discriminate between the associated probability distributions, which should not be necessarily a metric, although in this case it is).

5. Concluding remarks

This paper has attempted to present an updated and well-argued review on a statistical methodology to analyze fuzzy numbers data on the basis of a versatile metric.

On the basis of the methodology above one can get several consistent techniques for testing about the distributions of real-valued random variables (namely, goodness of fit tests, homogeneity tests, distribution estimators, etc.). This is based on the fact that there exist characterizing fuzzy representations of real-valued random variables leading to random fuzzy numbers. In González-Rodríguez et al. [27], some generalized characterizing fuzzy representations of random variables have been presented. The idea behind these representations consists of converting each (real-valued) variable value into a given fuzzy number in such a way that the Aumann-type mean value of the resulting random fuzzy number is a functional characterization of the distribution of the random variable. Intuitively, the characterizing ability has been achieved by basing the fuzzy conversion either on different polygonal functions with 'sharp points' (see Colubi et al. [4,5]) or on different 'smooth curvatures' (see González-Rodríguez et al. [27]) for different values. The derived tests and estimators are quite competitive because of being based on (generalized) mean values and consistent methods. Furthermore, one can also derive asymmetry coefficients and tests about the symmetry of a random variable, like the ones in González-Rodríguez et al. [24,29].

Acknowledgements

The research in this communication has been partially supported by the Grants from the Spanish Ministry of Education and Science MTM2009-09440-C02-01, the Principality of Asturias Grant SV-PA-13-ECOEMP-66, the Integrated Action Spain–Germany PRI-AIBDE-2011-1197, as well as the FPU Grant AP2009-1197 (Sinova), the FPI Grant BES-2010-032172 (García-Bárcana) and the Severo Ochoa-Principality of Asturias Grant BP12012 (de la Rosa de Sáa). Their financial support is gratefully acknowledged. The authors wish to thank the Guest Editors of the Special Issue for giving them the opportunity of attending to and participating (Ana Colubi) in the *Workshop on Harnessing the Information contained in low-quality data sources*, as well as to prepare this review for the International Journal of Approximate Reasoning. Some comments and remarks in this paper are in fact the answer to some insightful questions posed along the Workshop. Thanks should also be given to the four reviewers of the manuscript and 1st revision for their valuable comments and suggestions.

References

- [1] C. Bertoluzza, N. Corral, A. Salas, On a new class of distances between fuzzy numbers, *Mathw. Soft Comput.* 2 (1995) 71–84.
- [2] A. Blanco-Fernández, M.R. Casals, A. Colubi, N. Corral, S. de la Rosa de Saa, M. García-Bárcana, M.A. Gil, G. González-Rodríguez, M.T. López, M.A. Lubiano, M. Montenegro, A.B. Ramos-Guajardo, B. Sinova, Random fuzzy sets: a mathematical tool to develop statistical fuzzy data analysis, *Iranian J. Fuzzy Syst.* 10 (2) (2013) 1–28.
- [3] A. Colubi, R. Coppi, P. D'Urso, M.A. Gil, Statistics with fuzzy random variables, *Metron LXV* (3) (2007) 277–303.
- [4] A. Colubi, G. González-Rodríguez, M.A. Gil, A new characterization of discrete random variables by means of fuzzy sets: graphical features, in: *Abst. 56th Ses. Int. Stat. Inst., ISI 2007, Lisbon, Portugal, 2007*, p. 165.
- [5] A. Colubi, G. González-Rodríguez, M.A. Gil, M.R. Casals, A characterizing functional representation of continuous distributions focussed on relevant parameters, in: *Abst. Kick-off Meeting ERCIM WG on Comput. & Stat. and Int. Work. Comput. & Fin. Econom., Geneva, Switzerland, 2007*, p. 40.
- [6] A. Colubi, G. González-Rodríguez, M.A. Gil, W. Trutschnig, Nonparametric criteria for supervised classification of fuzzy data, *Int. J. Approx. Reason.* 52 (2011) 1272–1282.
- [7] I. Couso, L. Sánchez, Mark-recapture techniques in statistical tests for imprecise data, *Int. J. Approx. Reason.* 52 (2) (2011) 240–260.
- [8] T. Denœux, Maximum likelihood estimation from fuzzy data using the EM algorithm, *Fuzzy Sets Syst.* 183 (2011) 72–91.
- [9] T. Denœux, M.H. Masson, P.A. Hébert, Nonparametric rank-based statistics and significance tests for fuzzy data, *Fuzzy Sets Syst.* 153 (2005) 1–28.
- [10] P. Diamond, P. Kloeden, Metric spaces of fuzzy sets, *Fuzzy Sets Syst.* 100 (1999) 63–71.
- [11] N.R. Draper, W.G. Hunter, Transformations: some examples revisited, *Technometrics* 11 (1969) 129–132.
- [12] D. Dubois, H. Prade, Gradual elements in a fuzzy set, *Soft Comput.* 12 (2008) 165–175.
- [13] D. Dubois, H. Prade, Gradualness, uncertainty and bipolarity: Making sense of fuzzy sets, *Fuzzy Sets Syst.* 192 (2012) 3–24.
- [14] R. Féron, Ensembles aléatoires flous, *C. R. Math. Acad. Sci. Paris A* 282 (1976) 903–906.
- [15] R. Féron, Ensembles flous attachés à un ensemble aléatoire flou, *Publ. Econom.* 9 (1976) 51–66.
- [16] R. Féron, Sur les notions de distance et d'écart dans une structure floue et leurs applications aux ensembles aléatoires flous, *C. R. Math. Acad. Sci. Paris A* 289 (1979) 35–38.
- [17] M.B. Ferraro, R. Coppi, G. González-Rodríguez, A. Colubi, A linear regression model for imprecise response, *Int. J. Approx. Reason.* 51 (7) (2010) 759–770.
- [18] M. Fréchet, Les éléments aléatoires de nature quelconque dans un espace distancié, *Ann. Inst. Henri Poincaré* 10 (1948) 215–310.
- [19] M.A. Gil, A. Colubi, P. Terán, Random fuzzy sets: why, when, how, *BEIO* 30 (2014), in press.
- [20] M.A. Gil, M.A. Lubiano, M. Montenegro, M.T. López, Least squares fitting of an affine function and strength of association for interval-valued data, *Metrika* 56 (2002) 97–111.
- [21] M.A. Gil, M. Montenegro, G. González-Rodríguez, A. Colubi, M.R. Casals, Bootstrap approach to the multi-sample test of means with imprecise data, *Comput. Stat. Data Anal.* 51 (2006) 148–162.
- [22] E. Giné, J. Zinn, Bootstrapping general empirical measures, *Ann. Probab.* 18 (1990) 851–869.
- [23] G. González-Rodríguez, A. Blanco, A. Colubi, M.A. Lubiano, Estimation of a simple linear regression model for fuzzy random variables, *Fuzzy Sets Syst.* 160 (2009) 357–370.
- [24] G. González-Rodríguez, A. Colubi, P. D'Urso, P. Giordani, An asymptotic test about the asymmetry of random variables based on fuzzy tools, in: J. Lawry, E. Miranda, A. Bugarin, S. Li, M.A. Gil, P. Grzegorzewski, O. Hryniewicz (Eds.), *Soft Methods for Integrated Uncertainty Modelling*, Springer, Berlin/Heidelberg, 2006, pp. 87–94.
- [25] G. González-Rodríguez, A. Colubi, P. D'Urso, M. Montenegro, Multi-sample test-based clustering for fuzzy random variables, *Int. J. Approx. Reason.* 50 (2009) 721–731.
- [26] G. González-Rodríguez, A. Colubi, M.A. Gil, Fuzzy data treated as functional data. A one-way ANOVA test approach, *Comput. Stat. Data Anal.* 56 (2012) 943–955.
- [27] G. González-Rodríguez, A. Colubi, M.A. Gil, A fuzzy representation of random variables: an operational tool in exploratory analysis and hypothesis testing, *Comput. Stat. Data Anal.* 51 (2006) 163–176.
- [28] G. González-Rodríguez, A. Colubi, M.A. Gil, P. D'Urso, An asymptotic two dependent samples test of equality of means of fuzzy random variables, in: *Proc. COMPSTAT'2006, 2006*, <http://www.stat.unipg.it/iase/Proceedings/2006/COMPSTAT/CD/145.pdf>.
- [29] G. González-Rodríguez, A. Colubi, M.A. Gil, M.A. Lubiano, A new way of quantifying the symmetry of a random variable: estimation and hypothesis testing, *J. Stat. Plan. Inference* 142 (2012) 3061–3072.
- [30] G. González-Rodríguez, M. Montenegro, A. Colubi, M.A. Gil, Bootstrap techniques and fuzzy random variables: synergy in hypothesis testing with fuzzy data, *Fuzzy Sets Syst.* 157 (2006) 2608–2613.
- [31] F. Herrera, E. Herrera-Viedma, L. Martínez, A fuzzy linguistic methodology to deal with unbalanced linguistic term sets, *IEEE Trans. Fuzzy Syst.* 16 (2008) 354–370.
- [32] W. Hoeffding, A class of statistics with asymptotically normal distribution, *Ann. Math. Stat.* 19 (1948) 293–325.
- [33] E.P. Klement, M.L. Puri, D.A. Ralescu, Limit theorems for fuzzy random variables, *Proc. R. Soc. Lond. Ser. A, Math. Phys. Sci.* 407 (1986) 171–182.
- [34] R. Körner, Linear models with random fuzzy variables, PhD thesis, Technische Universität Bergakademie Freiberg, 1997, <http://www.mathe.tu-freiberg.de/inst/stoch/Koerner/dissa.ps>.
- [35] R. Körner, An asymptotic α -test for the expectation of random fuzzy variables, *J. Stat. Plan. Inference* 83 (2000) 331–346.
- [36] R. Körner, W. Näther, On the variance of random fuzzy variables, in: C. Bertoluzza, M.A. Gil, D.A. Ralescu (Eds.), *Statistical Modeling, Analysis and Management of Fuzzy Data*, Physica-Verlag, Heidelberg, 2002, pp. 22–39.
- [37] R. Kruse, K.D. Meyer, *Statistics with Vague Data*, D. Reidel Publishing Company, Dordrecht, 1987.
- [38] H. Levene, Robust tests for equality of variances, in: I. Olkin (Ed.), *Contributions to Probability and Statistics*, Stanford University Press, Palo Alto, 1960, pp. 278–292.
- [39] M.A. Lubiano, M.C. Alonso, M.A. Gil, Statistical inferences on the S-mean squared dispersion of a fuzzy random variable, in: *Proc. Joint EUROFUSE-SIC'99, 1999*, pp. 532–537.
- [40] M.A. Lubiano, M.A. Gil, M. López-Díaz, M.T. López, The $\bar{\lambda}$ -mean squared dispersion associated with a fuzzy random variable, *Fuzzy Sets Syst.* 111 (2000) 307–317.
- [41] M. Montenegro, M.R. Casals, M.A. Lubiano, M.A. Gil, Two-sample hypothesis tests of means of a fuzzy random variable, *Inform. Sci.* 133 (2001) 89–100.
- [42] M. Montenegro, A. Colubi, M.R. Casals, M.A. Gil, Asymptotic and bootstrap techniques for testing the expected value of a fuzzy random variable, *Metrika* 59 (2004) 31–49.
- [43] W. Näther, Random fuzzy variables of second order and applications to statistical inference, *Inform. Sci.* 133 (2001) 69–88.
- [44] H.T. Nguyen, A note on the extension principle for fuzzy sets, *J. Math. Anal. Appl.* 64 (1978) 369–380.
- [45] M.L. Puri, D.A. Ralescu, The concept of normality for fuzzy random variables, *Ann. Probab.* 11 (1985) 1373–1379.
- [46] M.L. Puri, D.A. Ralescu, Random fuzzy sets, *J. Math. Anal. Appl.* 114 (1986) 409–422.
- [47] A.B. Ramos-Guajardo, A. Colubi, G. González-Rodríguez, M.A. Gil, One-sample tests for a generalized Fréchet variance of a fuzzy random variable, *Metrika* 71 (2010) 185–202.

- [48] A.B. Ramos-Guajardo, G. González-Rodríguez, M. Montenegro, M.T. López, Power analysis of the homoscedasticity test for random fuzzy sets, in: C. Borgelt, G. González-Rodríguez, W. Trutschnig, M.A. Lubiano, M.A. Gil, P. Grzegorzewski, O. Hryniewicz (Eds.), *Combining Soft Computing and Statistical Methods in Data Analysis*, Springer-Verlag, Heidelberg, 2010, pp. 537–544.
- [49] A.B. Ramos-Guajardo, M.A. Lubiano, K -sample tests for equality of variances of random fuzzy sets, *Comput. Stat. Data Anal.* 56 (2012) 956–966.
- [50] W. Trutschnig, G. González-Rodríguez, A. Colubi, M.A. Gil, A new family of metrics for compact, convex (fuzzy) sets based on a generalized concept of mid and spread, *Inform. Sci.* 179 (2009) 3964–3972.
- [51] L.A. Zadeh, The concept of a linguistic variable and its application to approximate reasoning, Part 1, *Inform. Sci.* 8 (1975) 199–249;
L.A. Zadeh, The concept of a linguistic variable and its application to approximate reasoning, Part 2, *Inform. Sci.* 8 (1975) 301–353;
L.A. Zadeh, The concept of a linguistic variable and its application to approximate reasoning, Part 3, *Inform. Sci.* 9 (1975) 43–80.